DAY 5 OF 30

# Lakehouse or Warehouse?

The week-one question.
A practical way to decide.

Spark | T-SQL | Skills

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THE PROBLEM

# Two items that look like they do the same job.

| | | |
|---|---|---|
| 1 | **Both** | Store tables in OneLake |
| 2 | **Both** | Use the Delta format |
| 3 | **Both** | Can be queried with SQL |

So teams argue about features, when **the real difference is who writes the data, and how**.

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THE SIMPLE VERSION

# A workshop and a showroom.

The **workshop** takes raw material in any shape. You need tools and some skill, and you can build anything.

The **showroom** is orderly. Everything is finished, labelled and easy to browse.

| | |
|---|---|
| ✓ **Lakehouse** | The workshop. Spark and files |
| ✓ **Warehouse** | The showroom. SQL all the way |
| ✓ **OneLake** | The same floor under both |

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SIDE BY SIDE

# The real differences

1. **Write with**: Lakehouse uses Spark. Warehouse uses T-SQL
2. **Data types**: Lakehouse takes files. Warehouse takes tables
3. **Transactions**: Warehouse supports multi-table
4. **Skills**: Python and Spark, or SQL
5. **Storage**: both are Delta in OneLake

**Not either-or**

A warehouse can query lakehouse tables in the same workspace. Many teams use both.

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FOR ENGINEERS

# How I would decide

### Start from the team, not the tool

A SQL team will be productive in a Warehouse on day one. A Spark team will not want to leave notebooks.

### Look at the raw data

Files, JSON, images and large volumes point to a Lakehouse for the early layers.

### Check the write pattern

Stored procedures and multi-table updates in T-SQL point to a Warehouse.

### Combine them

Lakehouse for Bronze and Silver, Warehouse for Gold is a common, sensible split.

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THE HONEST PART

# Which one first?

### LAKEHOUSE FIRST

Engineers who work in Spark

Raw, messy or very large data

Data science on the same data

### WAREHOUSE FIRST

Analysts and developers who live in SQL

A migration from a SQL warehouse

Heavy use of stored procedures

> **The catch:** picking by feature list leads to rework. Pick by who will maintain it at 2 a.m., because that is who needs to read the code.

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FROM REAL WORK

# The choice on a real project

A manufacturing group with years of reporting logic in legacy SQL systems.

1. **Legacy SQL data** moved to Azure Synapse and Fabric
2. **Enterprise reporting** was centralised in one place
3. **Data Factory and gateways** connected on-premises sources
4. **PySpark** automated the heavier reporting pipeline

SQL where the logic already lived, **Spark where the heavy lifting was**.

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SAVE THIS

# The decision in five lines

1. **Both store Delta** in OneLake
2. **Lakehouse**: Spark writes, files welcome
3. **Warehouse**: full T-SQL, transactions
4. **Choose by skills** and write pattern
5. **Using both** is normal, not a compromise

## Tomorrow: Delta tables.
## The format under every table in Fabric.

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